output and the true output, and our goal is to find the set of parameters that minimizes this error function. There are many possible ways to minimize this function, not all of which depend on using derivatives, but in most cases the most effective and efficient way to optimize loss functions in machine learning is to use derivative information.

Since deep learning models are nonlinear (i.e., they do not preserve addition and scalar multiplication), the derivatives are not constant like in linear transformations. The amount and direction of squishing or stretching that happens from input to output points varies from point to point. In another sense, it tells us which direction the function is curving, so we can follow the curve downward to the lowest point. Multivariable functions like deep learning models don't just have a single derivative but a set of partial derivatives that describe the curvature of the function with respect to each individual input component. This way we can figure out which sets of parameters for a deep neural network lead to the smallest error.

The simplest example of using derivative information to minimize a function is to see how it works for a simple compositional function. The function we will try to find the minimum of is:

$$f(x) = \log(x^4 + x^3 + 2)$$

The graph is shown in figure A.1. You can see that the minimum of this function appears to be around –1. This is a compositional function because it contains a polynomial expression "wrapped" in a logarithm, so we need to use the chain rule from calculus to compute the derivative. We want the derivative of this function with respect to *x*. This function only has one "valley," so it will only have one minimum; however, deep learning models are high-dimensional and highly compositional and tend to have many minima. Ideally, we'd like to find the global minimum that is the lowest point in the function. Global or local minima are points on the function where the slope (i.e., the derivative) at those points is 0. For some functions, like this simple example, we can compute the minimum analytically, using algebra. Deep learning models are generally too complex for algebraic calculations, and we must use iterative techniques.